Lesson 12: Hearing It Happen

Scratch

VIDEO GAME CREATION

This week: your game finally sounds like a game. Four moments - firing, hitting, clearing a wave, and losing - each get their own distinct cue.

Open your own [YourName]Game project from Lesson 11 - don't start a new one.

If you already built an optional sound idea in an earlier lesson (Lessons 2, 3, 4, or 6), keep it - today reuses it directly instead of replacing it. If you didn't, today's steps are written as if starting fresh.

Setup

1. Open your [YourName]Game project from Lesson 11.
2. No new variables needed - today only adds sound blocks to code you already have.
3. Confirm sfx_wave is uploaded to GameManager's Sounds tab (today's one genuinely new sound asset) and sfx_hit is uploaded to the Enemy sprite's (Scratch sound assets are scoped per-sprite, so upload each to whichever sprite's script will play it). If you skipped Lesson 8's optional sound idea, also upload sfx_gameover from the Lesson 8 Assets folder to the Player sprite - it becomes mandatory starting today.

Level 1 - Minimal (MVP)

Four sounds fire at the correct moment, at least once each: a shot, a hit/explosion, a wave-clear (on any non-final wave), and a game over. No silence, no obviously wrong triggers.

4. On the Player sprite's space-key script, confirm "start sound [sfx_shoot]" already fires next to "create clone of [Bullet]" - if it's missing (you may have skipped this as an optional idea back in Lesson 3), add it now.
5. On the Enemy sprite's touching-Bullet branch, add "start sound [sfx_hit]" right before "broadcast [enemy hit]" - this plays once per hit, on the exact clone that was touched. Enemy never touches score or enemies_remaining itself (Rule 13) - that stays on GameManager.
6. On GameManager's "when I receive [enemy hit]" script, add "start sound [sfx_wave]" to the else branch (the wave-advance case), right before "broadcast [spawn enemies]".
7. On the Player sprite's "when I receive [game over]" script, confirm "start sound [sfx_gameover]" already fires (optional idea from Lesson 8) - add it now if it's missing, right before the "say" block. This sound is mandatory starting today even if you skipped it before.

8. Click the stage and play through a full round - shooting, hitting, an early wave clear, and losing a life to zero should each sound distinct.

Highlighted lines are new this week; everything else is code you already have.

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Enemy sprite's touching-Bullet branch (revised):
if <touching Bullet?> then
  start sound [sfx_hit]
  broadcast [enemy hit]
delete this clone

GameManager's "when I receive [enemy hit]" script (revised):
change enemies_remaining by -1
change score by 10
if <enemies_remaining = 0> then
  if <wave = total_waves> then
    set game_active to 0
    broadcast [you win]
  else
    change wave by 1
    set enemies_remaining to 5
    start sound [sfx_wave]
    broadcast [spawn enemies]

Player sprite's "when I receive [game over]" script (revised):
start sound [sfx_gameover]        // your Lesson 8 Level 3 sound,
mandatory as of today
say [Game Over - Press R to Restart] for 3 seconds
```

Level 2 - Complete (Alpha/Beta)

All four sounds fire reliably every time, with no missing or duplicate playback, and each is clearly distinguishable from the others by ear.

- Play a full round: shoot, hit an enemy, clear wave 1 or 2, and separately let lives hit 0 - all four sounds should be clearly different from each other.
- If you built Lesson 6's optional victory sound, confirm it still plays on the true win (wave 3's clear) and doesn't also play the wave-clear sound on the same frame.
- Rapid shooting and a hit landing in the same instant should layer, not cut each other off - "start sound" doesn't block or loop, so this should already work correctly.

Level 3 - Refined (Ready to Publish) - pick any that interest you

- Audio: if you haven't already, add Lesson 6's optional win sound now - "start sound [sfx_win]" right where "broadcast [you win]" fires on GameManager. It slots into the exact same branch as today's wave-clear sound, just the opposite outcome.
- Audio: design a second, alternate hit sound and randomly pick between the two each time (pick random 1 to 2), so repeated hits don't sound completely identical.
- Structure: notice all four sounds live in the exact spot representing their event - shoot with the shot, hit with the hit (on Enemy), wave/win with the branch that decides them (on GameManager), lose with the Player's own game-over script. None of them live in a generic "forever" script, and none of them require Enemy to know about score, wave, or lives.

If Something's Wrong

- No sound at all: confirm the correct sfx_ file was uploaded via the Sounds tab for the sprite the "start sound" block lives on - Scratch sound assets are scoped per-sprite, not shared automatically.
- Wave-clear and win sounds both play on wave 3: confirm "start sound [sfx_wave]" is only inside the else branch, not before the if/else splits.
- Game-over sound plays repeatedly, or doesn't play at all: confirm it lives on the Player's "when I receive [game over]" script (fires once per broadcast), not inside GameManager's per-frame checks.

Want More?

Student Resource List > Scratch section, or scratch.mit.edu/ideas